PROJECT REPORT

On

"E-Pharmacy Portal"

Submitted in partial fulfillment of the requirements for the award of Degree of

Masters of Computer Application

IN

DEPARTMENT OF COMPUTER APPLICATIONS

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2. Candidate’s Declaration

I,Kamar Raza, hereby declare that the Industrial Project Reportentitled “E-pharmacy Prescription” submitted to the Department of Computer Application, Engineering College Bikaner, affiliated to Bikaner Technical University (BTU), Rajasthan, in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications, is a record of original work carried out by me under the guidance of Dr. Abdul Jabbar Khilji MCA Department.

I further declare that this report has not been submitted to any other university or institution for the award of any degree or diploma.

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CHAPTER 1

INTRODUCTION

1.1 Organization Profile

The present project **“Design and Development of an E-Pharmacy Portal”** has been developed to support the digital transformation of healthcare services. The organization under which this project has been carried out is focused on providing **technology-enabled healthcare solutions** that connect patients, pharmacies, and service providers through a centralized web-based platform.

The organization aims to simplify traditional pharmacy operations by introducing automation, transparency, and faster communication between patients and medical stores. With the rapid growth of internet usage and digital payments, the healthcare sector is moving towards online platforms to improve service quality and accessibility. The E-Pharmacy Portal is a step in this direction.

The organization mainly works in the following domains:

- Web-based healthcare applications
- Digital prescription management
- Online order processing systems
- Secure data handling and storage

The project has been designed considering real-world pharmacy operations, legal requirements for prescription handling, and the convenience of both patients and pharmacy owners. The system ensures that only **authorized and subscribed pharmacies** can receive and process prescriptions uploaded by patients.

1.2 Existing System and Need for the System

Existing System

In the existing system, pharmacy operations are largely **manual and semi-digital**. Patients are required to physically visit nearby medical stores to submit their prescriptions and enquire about medicine availability. In many cases, patients have to visit multiple stores before finding all the required medicines.

The current system involves:

- Physical submission of prescriptions
- Manual checking of medicine availability
- Paper-based billing
- Cash-based or limited digital payment options

Problems in the Existing System

The existing system suffers from several limitations:

- **Time-consuming process** for patients
- **No centralized prescription tracking**
- **High dependency on physical presence**
- **Manual record maintenance** leading to errors
- **Limited access for elderly and critical patients**
- **No real-time confirmation of medicine availability**

Need for the Proposed System

With increasing demand for fast and reliable healthcare services, there is a strong need for an **online and automated pharmacy system**. The proposed E-Pharmacy Portal addresses the shortcomings of the existing system by enabling digital prescription upload, pharmacy verification, and online payments.

The system helps:

- Patients to save time and effort
- Pharmacies to manage prescriptions efficiently
- Administrators to monitor and control operations

1.3 Scope of Work

The scope of the **E-Pharmacy Portal** defines the boundaries and functionality of the proposed system. The project focuses on developing a **secure, user-friendly, and scalable web application**.

In-Scope Features

- User registration and login (Patient & Pharmacy)
- Subscription-based pharmacy onboarding
- Prescription upload by patients
- Prescription review by nearby pharmacies
- Acceptance or rejection of prescriptions
- Medicine availability confirmation

- Online payment integration
- Order status tracking
- Admin panel for system management

Out-of-Scope Features

- Manufacturing of medicines
- Medical diagnosis or consultation
- Emergency services

The system is designed in such a way that it can be enhanced in the future with mobile applications, GPS-based pharmacy search, and AI-based medicine recommendations.

1.4 Operating Environment – Hardware and Software

Hardware Requirements

The following hardware configuration is sufficient for developing and running the E-Pharmacy Portal:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Hard Disk:** 500 GB
- **Network:** Broadband Internet Connection

Software Requirements

System Software

- Operating System: Windows 10 / Linux
- Web Server: Apache / IIS

Application Software

-
- Front-End Technologies:
  - HTML
  - CSS
  - JavaScript
- Back-End Technologies:
  - Node.js
- Database:

    - MySQL

- Development Tools:

    - Visual Studio Code

User Environment

The application can be accessed through:

- Desktop Computers
- Laptops
- Tablets
- Smartphones with a web browser

1.5 Problem Definition

Add a separate section explaining the exact problem:

- Patients waste time searching medicines
- No centralized prescription management
- Lack of online pharmacy coordination
- Difficulty for elderly and emergency patients

1.6 Project Motivation

Explain why this project was developed:

- Growth of digital healthcare
- Increase in online services
- Need for contactless medicine ordering
- Better healthcare accessibility

1.7 Project Goals

Add measurable goals:

- Reduce medicine search time
- Improve pharmacy response time
- Maintain secure digital records
- Enable paperless workflow

1.5 Technologies Used in the Project

The E-Pharmacy Portal is developed using modern web technologies to provide a secure, scalable, responsive, and user-friendly healthcare platform. The project follows a full-stack web application architecture consisting of frontend technologies, backend technologies, database systems, authentication mechanisms, APIs, and cloud/database services.

Frontend Technologies

The frontend of the application is responsible for the user interface and user experience. It allows patients, pharmacies, and administrators to interact with the system through a responsive web interface.

1. Next.js 16

Next.js is used as the primary frontend framework for building the web application. Next.js provides server-side rendering, routing, API handling, and optimized performance for modern web applications.

Features used in the project:

- File-based routing
- Dynamic pages
- API route integration
- Fast page rendering
- Optimized frontend performance
- Responsive UI support

---

2. React.js

React is used for creating reusable user interface components and interactive pages.

React helps in:

- Component-based architecture
- Faster UI updates
- State management
- Better user experience

---

3. HTML, CSS, and JavaScript

Basic frontend technologies used include:

- HTML for page structure
- CSS for styling and responsiveness
- JavaScript for interactivity and dynamic behavior

These technologies help create a modern and mobile-friendly healthcare portal.

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4. Google Maps API

Google Maps Platform is integrated into the project for location-based pharmacy discovery.

Features include:

- Nearby pharmacy search
- Map integration
- Location selection
- Real-time pharmacy identification

This feature helps patients quickly locate nearby subscribed pharmacies.

Backend Technologies

The backend handles business logic, authentication, prescription processing, order management, and database communication.

---

1. Next.js API Routes

The backend APIs are developed using Next.js API Routes located inside:

app/api/.../route.ts

These API routes handle:

- User authentication
- Prescription upload
- Pharmacy management
- Payment processing
- Database operations

---

2. Node.js Runtime

Node.js is used as the backend runtime environment through Next.js.

Node.js provides:

- Fast server-side execution
- Asynchronous processing
- Efficient API handling
- Scalable backend architecture

---

3. MongoDB Database

MongoDB is used as the primary database system for storing application data.

MongoDB stores:

- Patient details
- Pharmacy records
- Prescriptions
- Orders
- Payments
- Subscription information

Advantages of MongoDB:

- Flexible schema
- High scalability
- Fast query processing
- JSON-based document storage

---

4. Mongoose ODM

Mongoose is used as the Object Data Modeling (ODM) library for MongoDB.

Mongoose helps in:

- Schema validation
- Data modeling
- Database query handling
- Middleware support

Authentication and Security Technologies

Security is an important part of the E-Pharmacy Portal because it handles healthcare-related user information and prescription data.

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1. JWT Authentication

JSON Web Token authentication is implemented using the jsonwebtoken library.

JWT is used for:

- Secure user authentication
- Session verification
- Protected API access
- Role-based access control

---

2. Password Hashing

Passwords are securely encrypted using bcryptjs.

Benefits:

- Secure password storage
- Data protection
- Prevention of password theft

---

3. HTTP-only Cookies

The system uses HTTP-only cookies for managing login sessions securely.

Advantages:

- Prevents client-side JavaScript access
- Protects against token theft
- Improves session security

Database Hosting and Deployment

Docker Compose

Docker Compose is used to manage local development services.

The project includes:

- Local MongoDB service
- Containerized environment
- Simplified deployment setup

Example:

mongo:6.0

Docker helps maintain consistency between development and deployment environments.

Overall Technology Stack

Layer	Technology Used
Frontend	Next.js 16, React.js, HTML, CSS, JavaScript
Backend	Next.js API Routes, Node.js
Database	MongoDB
ODM	Mongoose
Authentication	JWT, bcryptjs
Session Management	HTTP-only Cookies
Maps & Location	Google Maps API
Containerization	Docker Compose
Database Hosting	MongoDB Atlas / Local MongoDB

1.9 Database Technology Used – MongoDB

MongoDB is a NoSQL document-oriented database used for storing large amounts of flexible and unstructured data. Unlike traditional relational databases, MongoDB stores data in JSON-like documents called BSON (Binary JSON), which makes data storage faster and more scalable.

In the E-Pharmacy Portal, MongoDB is used to manage dynamic healthcare-related data efficiently and support real-time application performance.

Why MongoDB Was Used

The E-Pharmacy Portal requires handling different types of user data, prescription files, order details, and pharmacy records. MongoDB was selected because of the following advantages:

- Flexible schema design
- Faster data retrieval
- Easy integration with Node.js
- Better scalability
- Efficient handling of JSON data
- High performance for web applications
- Cloud database support using MongoDB Atlas

Role of MongoDB in the Project

MongoDB is used as the primary database system for storing and managing application data such as:

- Patient information
- Pharmacy details
- Prescription records
- Order history
- Payment information
- Subscription plans
- Authentication details

The database communicates with the backend server using APIs and stores all records securely.

MongoDB Architecture Used

The project follows the following flow:

Client (Frontend)

↓

Node.js / Express Server

↓

MongoDB Database

↓

Data Response to User

The backend server processes user requests and interacts with MongoDB to perform CRUD operations.

MongoDB Collections Used

1. Patients Collection

Stores patient-related details.

Example fields:

- _id
- name
- email
- password
- phone
- address

---

2. Pharmacies Collection

Stores pharmacy registration and subscription details.

Example fields:

- pharmacy_name
- owner_name
- email
- subscription_status
- location

---

3. Prescriptions Collection

Stores uploaded prescription information.

Example fields:

- patient_id
- pharmacy_id
- prescription_image
- upload_date
- status

---

4. Orders Collection

Stores medicine order details.

Example fields:

- order_id
- prescription_id
- order_status
- total_amount

---

5. Payments Collection

Stores payment transaction records.

Example fields:

- payment_method
- amount
- payment_status

- transaction_id

Advantages of Using MongoDB in This Project

1. Flexible Data Structure

MongoDB allows storing different types of healthcare records without fixed table structures.

2. Faster Development

MongoDB works efficiently with JavaScript and Node.js, reducing development complexity.

3. Scalability

The system can support large numbers of users and pharmacies in future expansion.

4. Cloud Database Support

MongoDB Atlas allows cloud-based database hosting with secure remote access.

5. Better Performance

MongoDB provides fast query execution and efficient handling of large datasets.

MongoDB Atlas Usage

The project uses MongoDB Atlas for cloud database hosting.

Features used:

- Cloud storage
- Secure authentication
- Remote database access
- Automatic backups
- Database monitoring

MongoDB Connectivity

The backend application connects MongoDB using connection strings and database drivers.

Example technologies used:

- Mongoose
- Express.js
- Node.js

Mongoose ORM

Mongoose is used for database schema modeling and validation.

Benefits:

- Schema validation
- Easier CRUD operations
- Middleware support
- Relationship handling

Example MongoDB Schema

```
const PatientSchema = new mongoose.Schema({

 name: String,

 email: String,

 password: String,

 phone: String,

 address: String

});
```

Security Features in MongoDB

The following security practices are implemented:

- Encrypted passwords
- Secure database connection
- Authentication-based access
- Environment variable protection
- Role-based access control

CHAPTER 2

PROPOSED SYSTEM

2.1 Proposed System

The proposed system, titled **"E-Pharmacy Portal"**, is a **web-based healthcare application** designed to digitally connect patients and subscribed medical stores through a centralized online platform. The system aims to replace the traditional manual pharmacy process with an **automated, secure, and user-friendly digital solution**.

In this system, patients can upload their prescriptions online instead of visiting multiple medical stores physically. The uploaded prescription is forwarded to nearby **subscribed pharmacies**, where the pharmacist verifies the prescription and checks the availability of medicines. Based on availability, the pharmacy can either **accept or reject** the request.

Once the prescription is accepted, the patient receives a confirmation and can proceed with **online payment**. After successful payment, the order status is updated and the patient is informed about order completion or pickup details.

The system also includes an **admin module** that controls overall operations such as pharmacy registration, subscription management, user monitoring, and report generation.

The proposed system ensures:

- Faster processing of prescriptions
- Reduced manual work
- Secure handling of medical data
- Improved service quality

2.2 Objectives of the System

The primary objective of the E-Pharmacy Portal is to **automate pharmacy operations** and improve accessibility to medicines using digital technology.

Primary Objectives

- To provide an online platform for prescription submission
- To reduce the time and effort required to obtain medicines
- To eliminate unnecessary physical visits to medical stores
- To ensure secure and systematic prescription handling

Secondary Objectives

- To enable pharmacies to manage orders efficiently
- To provide real-time updates to patients
- To integrate online payment facilities
- To maintain accurate digital records
- To improve transparency and accountability

The system is designed keeping in mind both **user convenience** and **operational efficiency**.

2.3 User Requirements

The E-Pharmacy Portal is a **multi-user system**. Each user type has specific roles and requirements.

2.3.1 Admin Requirements

The admin is responsible for managing the entire system.

Admin Functions:

- Login authentication
- Manage patient accounts
- Approve and manage pharmacy registrations
- Monitor subscription status of pharmacies
- View and generate reports
- Handle system security and maintenance

2.3.2 Pharmacy Requirements

Pharmacies must be registered and subscribed to access the system.

Pharmacy Functions:

- Secure login

- View uploaded prescriptions
- Verify prescription authenticity
- Check medicine availability
- Accept or reject prescriptions
- Update order status
- View payment confirmation

2.3.3 Patient Requirements

Patients are the end users of the system.

Patient Functions:

- User registration and login
- Upload prescription images or documents
- View pharmacy responses
- Confirm orders
- Make online payments
- Track order status
- View order history

2.4 Advantages of the Proposed System

- Reduces time and effort for patients
- Improves efficiency of pharmacies
- Minimizes human errors
- Provides secure digital records
- Supports online payment systems
- Accessible anytime and anywhere

2.5 Feasibility Study

2.5.1 Technical Feasibility

The system uses standard web technologies such as HTML, CSS, JavaScript, and MySQL, which are easily available and supported.

2.5.2 Operational Feasibility

The system is user-friendly and requires minimal training for users.

2.5.3 Economic Feasibility

The project is cost-effective as it uses open-source tools and software.

2.6 Functional Requirements

Add detailed functional requirements like:

Patient Module

- Registration/Login
- Upload Prescription
- Payment Processing
- Order Tracking

Pharmacy Module

- Prescription Review
- Order Management
- Status Update

Admin Module

- User Management
- Report Generation
- Subscription Monitoring

2.7 Non-Functional Requirements

Very important for MCA reports.

Include:

- Security
- Scalability
- Performance
- Reliability
- Availability
- Usability
- Maintainability

Example:

> The system must respond within minimum processing time and support

multiple users simultaneously.

CHAPTER 3

ANALYSIS & DESIGN

3.1 Entity Relationship Diagram (ERD)

Entity Relationship Diagram (ERD) is used to represent the logical structure of the database and the relationship between different entities involved in the system.

Main Entities Identified

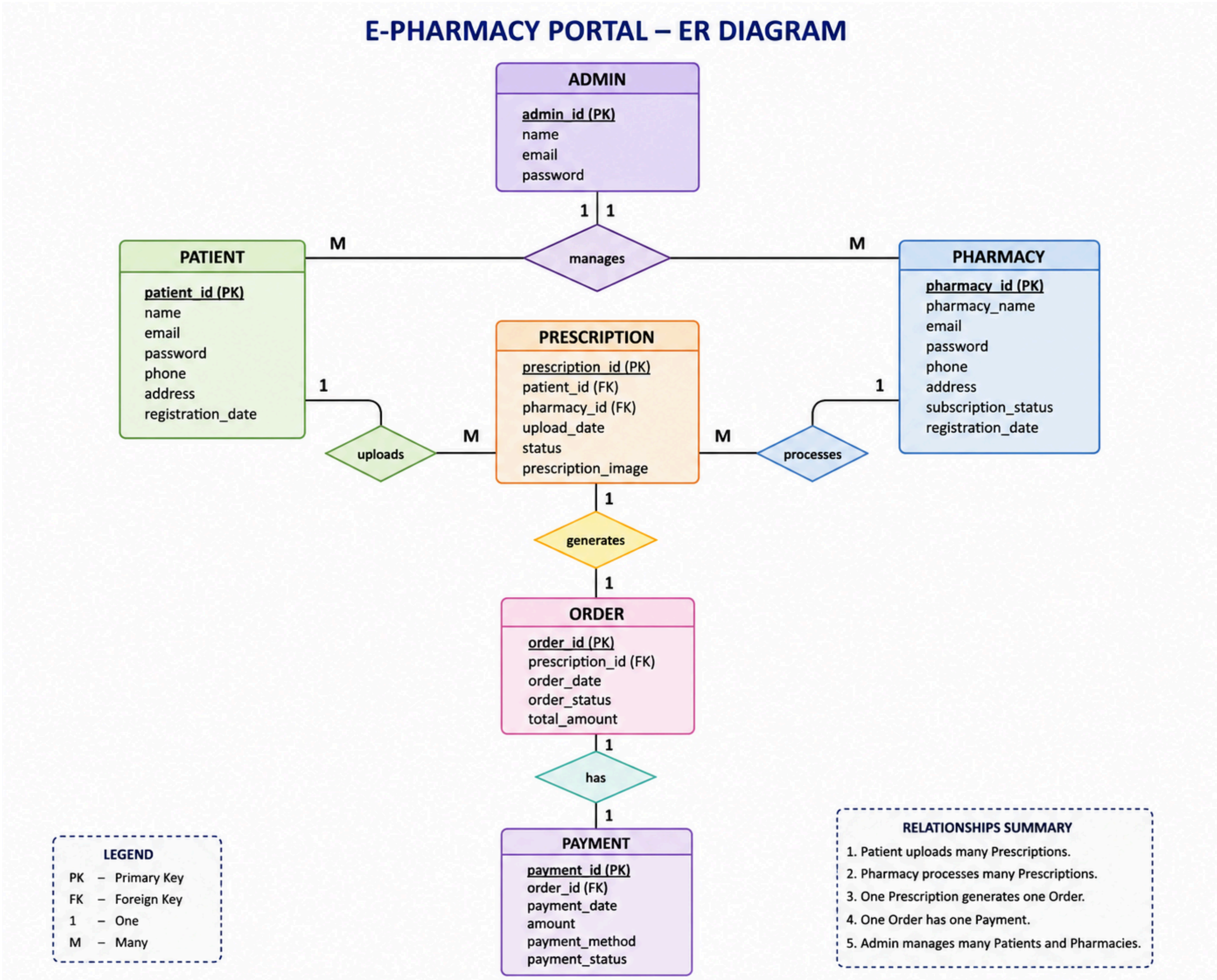
1. **Admin**
2. **Patient**
3. **Pharmacy**
4. **Prescription**
5. **Order**
6. **Payment**

Relationships

- A **Patient** can upload **multiple Prescriptions**
- A **Pharmacy** can receive **multiple Prescriptions**
- Each **Prescription** is linked to one **Order**
- Each **Order** has one **Payment**
- **Admin** manages Patients and Pharmacies

ERD Description

- Patient (Patient_ID) uploads Prescription (Prescription_ID)
- Pharmacy (Pharmacy_ID) processes Prescription
- Order (Order_ID) is generated after acceptance
- Payment (Payment_ID) is associated with Order



3.2 System Architecture

The E-Pharmacy Portal follows a **three-tier architecture**:

Presentation Layer

- User Interface (Web Browser)
- HTML, CSS, JavaScript

Application Layer

- Business Logic
- Prescription processing

- Order management
- Payment validation

Database Layer

- MySQL Database
- Stores user, prescription, order, and payment data

Architecture Flow:
Client → Web Server → Application Logic → Database → Response to Client

---

3.3 Database Requirements & User Interfaces

Database Requirements

The database must:

- Store user credentials securely
- Maintain prescription records
- Track order and payment details
- Ensure data integrity and consistency

User Interfaces

The system provides separate dashboards for:

- Admin
- Pharmacy
- Patient

Each interface is designed to be user-friendly and responsive.

3.4 Data Flow Diagram (DFD)

DFD Level 0

- Patient uploads prescription
- Pharmacy verifies prescription
- Admin monitors activities

DFD Level 1

1. User Authentication
2. Prescription Upload
3. Prescription Review
4. Order Processing
5. Payment Processing

3.5 Data Dictionary

Field Name	Data Type	Description
user_id	INT	Unique user identifier
name	VARCHAR	User name
email	VARCHAR	Email address
role	VARCHAR	User type
prescription_id	INT	Prescription number
order_id	INT	Order number
payment_id	INT	Payment number

3.6 Table Design

Patient Table

```
CREATE TABLE Patient (

patient_id INT PRIMARY KEY AUTO_INCREMENT,

name VARCHAR(50),

email VARCHAR(50),

password VARCHAR(100),

phone VARCHAR(15)

);
```

Pharmacy Table

```
CREATE TABLE Pharmacy (
pharmacy_id INT PRIMARY KEY AUTO_INCREMENT,
pharmacy_name VARCHAR(100),
email VARCHAR(50),
subscription_status VARCHAR(20)
);
```

Prescription Table

```
CREATE TABLE Prescription (
prescription_id INT PRIMARY KEY AUTO_INCREMENT,
patient_id INT,
pharmacy_id INT,
upload_date DATE,
status VARCHAR(20)
);
```

3.7 Code Design

The system is divided into modules for easy maintenance:

- Authentication Module
- Prescription Management Module
- Order Management Module
- Payment Module
- Admin Management Module

Each module consists of reusable functions and follows modular programming principles.

3.8 Menu Screens

Admin Menu

- Dashboard
- Manage Pharmacies
- View Reports

Patient Menu

- Upload Prescription
- Track Orders
- Payment History

Pharmacy Menu

- View Prescriptions
- Accept / Reject Orders
- Order History

3.9 Input Screens

- Login Form
- Registration Form
- Prescription Upload Form
- Payment Form

Each input screen includes validation to ensure correct data entry.

3.10 Report Formats

- Prescription Report
- Order Report
- Payment Report
- Pharmacy Subscription Report

Reports are generated in tabular format for easy understanding.

3.11 Test Procedures and Implementation

Test Cases

Test Case	Input	Expected Output	Result
Login	Valid Credentials	Login Success	Pass
Upload Prescription	Image File	Uploaded	Pass
Payment	Valid Details	Payment Success	Pass

Implementation

The system is implemented using web technologies and tested on multiple browsers to ensure compatibility and performance.

3.12 UML Diagrams

Add UML diagrams section.

Include:

- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Class Diagram

This increases report quality significantly.

3.13 Use Case Description

Example:

Actor	Function
Patient	Upload Prescription
Pharmacy	Accept/Reject Request
Admin	Monitor System

3.14 Input Validation

Explain validations:

- Email validation
- Password validation
- File upload restrictions
- Required field validation

3.15 Security Mechanisms

Add:

- Password Encryption
- Session Authentication
- SQL Injection Prevention
- Secure API Communication
- Role-Based Access Control

CHAPTER 4

USER MANUAL

4.1 Introduction to User Manual

This chapter describes the operational procedure of the “E-Pharmacy Portal”. The system is designed for three types of users:

1. Admin
2. Pharmacy (Subscribed Partner)
3. Patient

The portal provides a secure and user-friendly interface for prescription upload, pharmacy matching, order confirmation, and payment processing.

The application can be accessed through a web browser using the deployed URL or local server (e.g., http://localhost:3000).

4.2 Patient User Manual

4.2.1 Patient Registration

To use the system, a patient must first create an account.

Steps:

1. Open the homepage of the E-Pharmacy Portal.
2. Click on the “Register” button.
3. Select role as “Patient”.
4. Fill in the required details:
    - Full Name
    - Email
    - Password
    - Phone Number
    - Address
5. Click on “Register”.

After successful registration, the patient can log in using their email and password.

4.2.2 Patient Login

1. Click on “Login”.
2. Select role as “Patient”.
3. Enter Email and Password.
4. Click “Login”.

If credentials are valid, the user is redirected to the patient dashboard.

4.2.3 Uploading Prescription

The main feature of the system is prescription upload.

Steps:

1. Click on “Upload Prescription”.
2. Enter patient details (if not auto-filled):
    - Full Name
    - Email
    - Phone Number
    - Location (City/ZIP)
3. Upload prescription image (PNG/JPG/WebP).
4. Add optional notes (if required).
5. Click “Submit”.

After submission:

- The prescription status becomes “Pending”.
- Nearby subscribed pharmacies receive the request.

4.2.4 Tracking Prescription Status

The patient can track prescription status from:

- Dashboard
- History section

Possible status values:

- Pending
- Accepted
- Fulfillment Requested
- Completed

4.2.5 Payment Process

Once the pharmacy accepts the prescription:

1. Patient receives confirmation.
2. Click on “Proceed to Payment”.
3. Enter payment details.
4. Confirm transaction.

After successful payment:

- Order status updates automatically.
- Confirmation message is displayed.

4.3 Pharmacy User Manual

4.3.1 Pharmacy Registration

Pharmacies must subscribe before accessing prescription requests.

Steps:

1. Click on “Register”.
2. Select role as “Pharmacist”.
3. Enter:
    - Pharmacy Name
    - Email
    - Password
    - Phone Number
    - Address
4. Choose setup method:
    - Add on Map
      OR
    - Manual PIN entry
5. Select subscription plan:
    - Monthly
    - Yearly
    - Premium
6. Complete registration.

After approval and subscription activation, pharmacy can log in.

4.3.2 Pharmacy Login

1. Click “Login”.
2. Select role as “Pharmacist”.
3. Enter email and password.
4. Click “Login”.

User is redirected to the Pharmacy Dashboard.

4.3.3 Pharmacy Dashboard Overview

The dashboard displays:

- Live updates status
- Location filter (City/ZIP)
- Total Activity
- Open Requests
- Completed Orders
- Patients Visited Today
- Recent Activity Panel

4.3.4 Viewing Prescription Requests

When a patient uploads a prescription:

1. It appears under “Recent Activity”.
2. Click on the prescription card.
3. View details:
    - Patient Name
    - Contact Details
    - Location
    - Uploaded Image
    - Current Status

4.3.5 Accepting or Rejecting Prescription

Pharmacy has two options:

- Accept Prescription

- Reject Prescription

If accepted:

- Status changes to "Fulfillment Requested".
- Patient receives notification.

If rejected:

- Patient can search another pharmacy.

4.3.6 Order Completion

After medicine preparation:

1. Confirm order readiness.
2. Update status to "Completed".
3. Patient receives final notification.

4.5 Forms and Report Specifications

Input Forms

- Registration Form
- Login Form
- Prescription Upload Form
- Pharmacy Setup Form
- Payment Form

Each form includes:

- Field validation
- Required field check
- Proper error messages
- Secure data submission

Reports Generated

- Prescription Report

- Order Status Report
- Payment Report
- Pharmacy Subscription Report

Reports are displayed in tabular format and stored in database for future reference.

4.6 Security Features

- Role-based authentication
- Encrypted password storage
- Session management
- Secure file upload validation
- Controlled access dashboards

4.7 System Limitations

- Requires internet connectivity
- Limited to subscribed pharmacies
- Dependent on correct prescription upload

4.8 Future Enhancements

- Mobile Application (Android/iOS)
- AI-based medicine availability prediction
- SMS/WhatsApp notifications
- Live inventory sync with pharmacy software
- Multi-city expansion with GPS integration

4.9 Error Handling

Explain common errors:

- Invalid Login
- Unsupported File Upload
- Payment Failure
- Session Timeout

4.10 Troubleshooting

Problem	Solution
Login Failed	Check Email/Password
Upload Error	Upload valid image
Payment Error	Retry transaction

CHAPTER 5
SYSTEM TESTING

5.1 Testing Overview

Explain importance of testing.

5.2 Types of Testing

Add:

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

5.3 Performance Testing

Explain:

- Fast response time
- Multiple user handling
- Server efficiency

CHAPTER 6 –
RESULT & CONCLUSION

6.1 Result Analysis

Explain achieved outputs:

- Faster prescription processing
- Better pharmacy management
- Secure transactions

6.2 Advantages Achieved

- Reduced paperwork
- Time saving
- Improved healthcare access

6.3 Limitations

- Internet dependency
- Limited pharmacy coverage
- No live inventory sync
- No Home Delivery

CHAPTER 7
FUTURE SCOPE

Add advanced future features:

- AI-based medicine recommendation
- Video consultation
- GPS live tracking
- WhatsApp integration
- Mobile App
- Voice assistant support
- Multi-language support
- Doorstep Delivery

Technology Stack Table

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	Node.js/PHP
Database	MySQL
Payment	Razorpay/Stripe(soon)

User Manual

1. Modern Healthcare Access

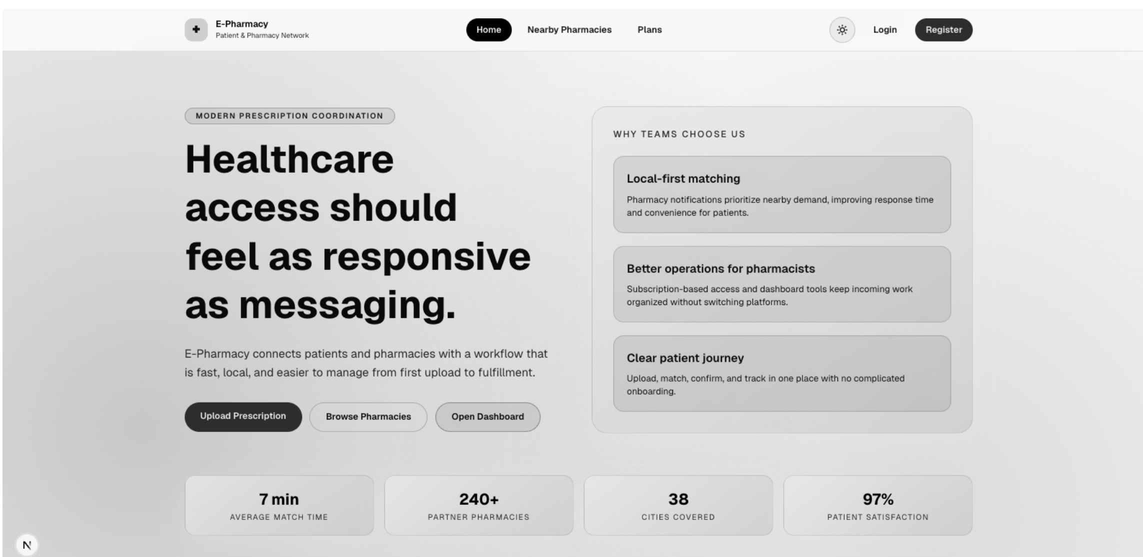
The homepage introduces the platform with a modern UI that highlights the mission of making healthcare access **as responsive as messaging**.

Main highlights include:

- Quick prescription uploads
- Access to nearby pharmacies
- Fast prescription fulfillment workflow
- Easy patient-pharmacy communication

The system shows platform statistics such as:

- Average match time
- Partner pharmacies
- Cities covered
- Patient satisfaction rate



2. Pharmacy Subscription Plans

The platform offers subscription tiers for pharmacies to access advanced features.

Monthly Plan – ₹299/month

- Access to prescriptions
- Basic dashboard
- Email support

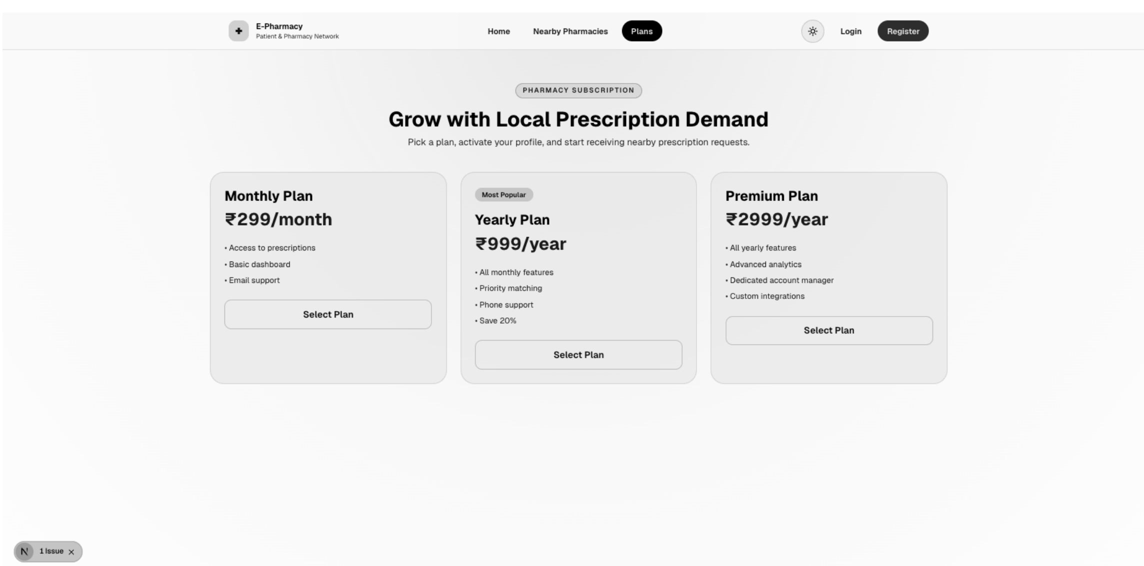
Yearly Plan – ₹999/year

- All monthly features
- Priority matching
- Phone support
- 20% savings

Premium Plan – ₹2999/year

- Advanced analytics
- Dedicated account manager
- Custom integrations

These plans help pharmacies manage prescription requests more efficiently while providing scalable service options.



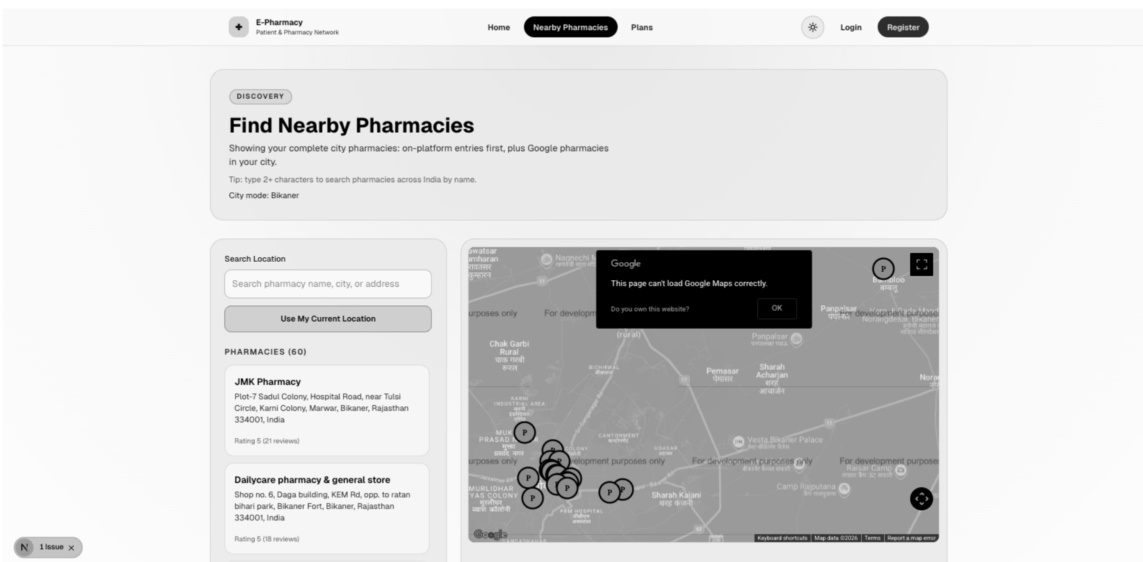
3. Nearby Pharmacy Discovery

Users can search for pharmacies based on their **city, ZIP code, or current location**.

Features include:

- Interactive Google Maps integration
- Search by pharmacy name, city, or address
- List of pharmacies with ratings
- Real-time location detection

This feature ensures patients can quickly find the **closest pharmacy available for prescription fulfillment**.



4. Secure User Authentication

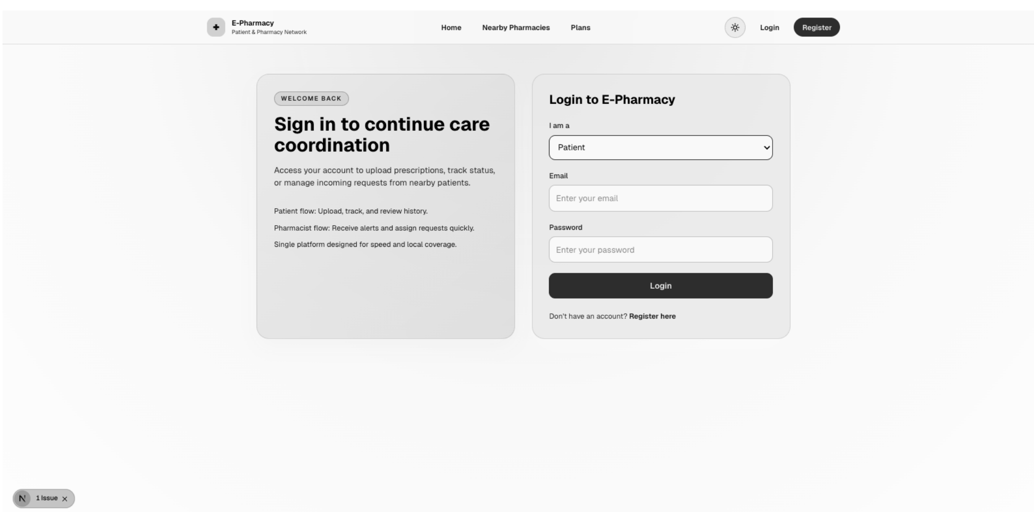
The platform provides a secure authentication system for both **patients and pharmacies**.

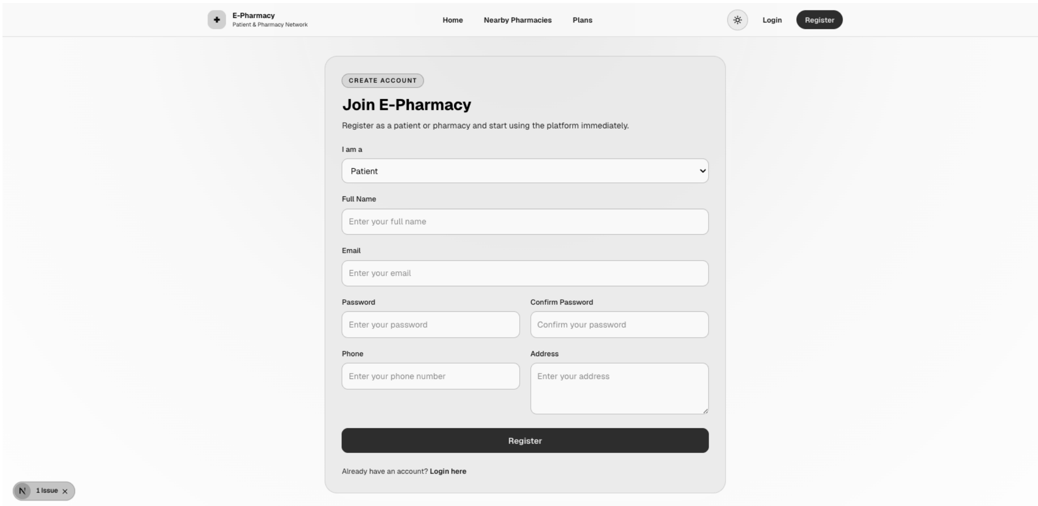
Users can:

- Register as a patient or pharmacy
- Login securely
- Manage their account information
- Access personalized dashboards

Registration requires:

- Name
- Email
- Password
- Phone number
- Address





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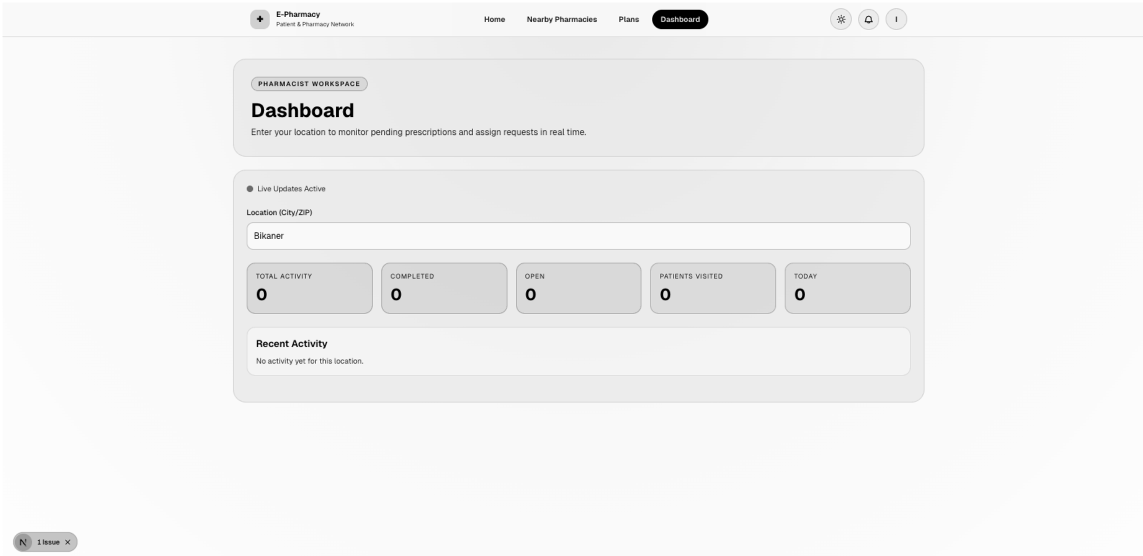
5. Pharmacy Dashboard

Pharmacies receive a **dashboard interface** to monitor and manage prescription requests.

Dashboard features include:

- Live activity updates
- Prescription request tracking
- Patient details
- Prescription images
- Status indicators (Open, Completed, Pending)

Pharmacists can **accept prescriptions and initiate fulfillment** directly from the dashboard.



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6. Prescription Upload System

Patients can securely upload their medical prescriptions through a dedicated upload page.

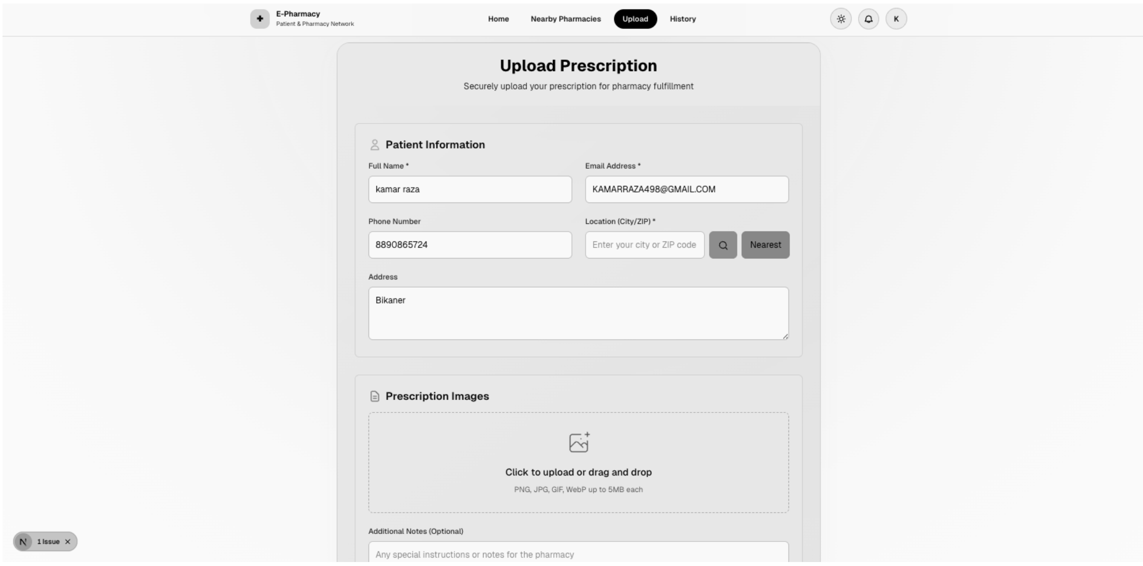
The form includes:

- Patient name
- Email
- Phone number
- Address
- City/ZIP code
- Prescription image upload

Supported formats include:

- PNG
- JPG
- GIF
- WebP

Uploaded prescriptions are sent to nearby pharmacies for quick response.



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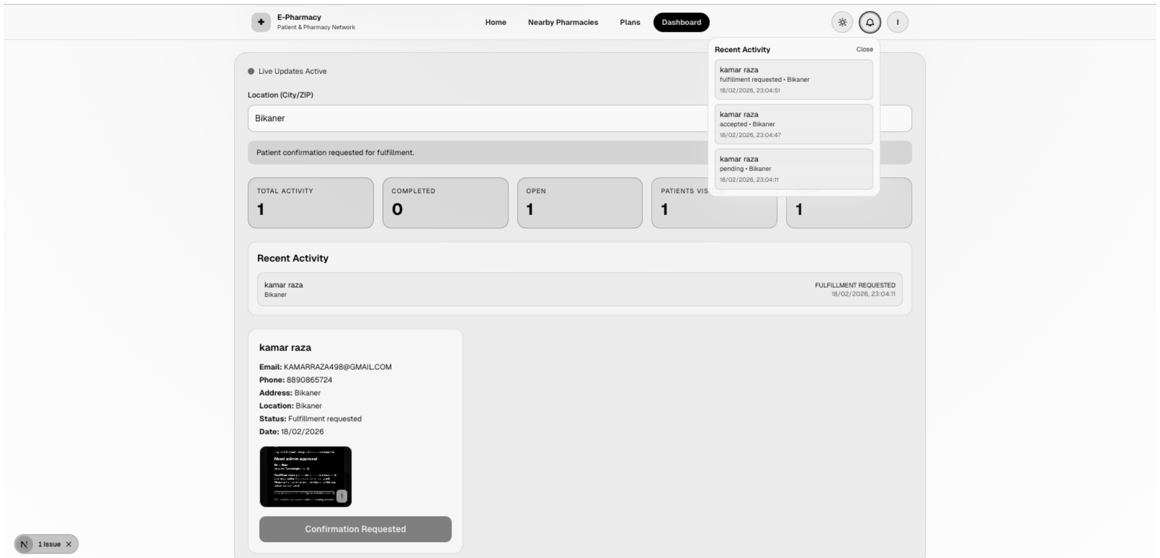
7. Activity Tracking

The system keeps a record of user actions such as:

- Prescription requests
- Fulfillment updates
- Confirmation requests
- Pharmacy responses

This ensures **transparency and easy monitoring of prescription status**.


E-Pharmacy
Patient & Pharmacy Network
Home
Nearby Pharmacies
Plans
Dashboard
Recent Activity
Close
kamar raza
fulfillment requested • Bikaner
18/02/2026, 23:04:51
kamar raza
accepted • Bikaner
18/02/2026, 23:04:47
kamar raza
pending • Bikaner
18/02/2026, 23:04:11
Live Updates Active
Location (City/ZIP)
Bikaner
Patient confirmation requested for fulfillment.
TOTAL ACTIVITY
1
COMPLETED
0
OPEN
1
PATIENTS VIS
1
1
Recent Activity
kamar raza
Bikaner
FULFILLMENT REQUESTED
18/02/2026, 23:04:11
kamar raza
Email: KAMARRAZA498@GMAIL.COM
Phone: 8890865724
Address: Bikaner
Location: Bikaner
Status: Fulfillment requested
Date: 18/02/2026
Confirmation Requested
1 Issue